

# Course

## « *Computer Vision* »

Sid-Ahmed Berrani

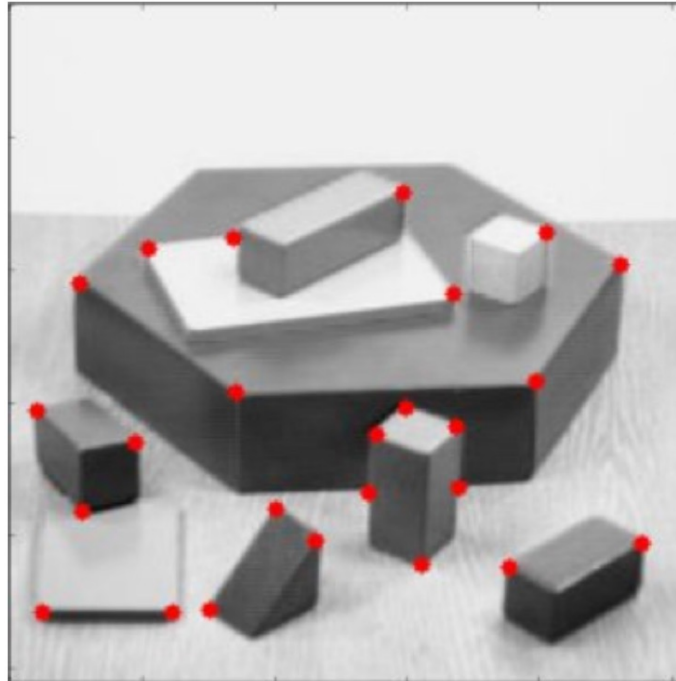
2025-2026



# Image Features

## What is a feature?

- Local, meaningful, detectable parts of the image.



# Image Features in Computer Vision

## **What is a feature?**

- Location of sudden change

## **Why use features?**

- Information content is high
- Invariant to change of viewpoint or illumination
- Reduces computational burden

# Image Features in Computer Vision

A good feature is **invariant** to:

- Viewpoint
- Lighting conditions
- Object deformations
- Partial occlusions

and should be:

- Unique
- Easy to be found and extracted

# Image Features: Edges

Edge pixels are pixels at which the intensity of an image function changes abruptly.

They represent a simple and meaningful type of image feature...

**why?**

- Edges in an image have many interesting causes
- Looking at edges reduces information required – Look at a few pixels in a binary image as opposed to all pixels in a grayscale image
- Biological plausibility – Initial stages of mammalian vision systems involve detection of edges and local features

# Image Features: Edges

*A sculpture*



Henry Moore sculpture – 1964.

<https://www.wikiart.org/fr/henry-moore/working-model-for-three-way-piece-no-2-archer-1964>

*A sketch made by an artist*



With a few strokes, the artist was able to convey a lot of information about the sculpture: The three-dimensional structure of the sculpture, some of the lighting effects like shading and highlights...

# Image Features: Focus on Edge Detection

## Objective:

Convert a 2D image into a set of points where intensity changes rapidly.

## Topics:

1. What is an edge?
2. Edge detection using gradients.
3. Edge detection using Laplacian.
4. Canny edge detector.
5. Line/curve boundary detection.
6. Hough transform.
7. Corner detection (the Haris detector).

# Image Features: Edges

- What causes an edge?



**Depth discontinuity**



# Image Features: Edges

- What causes an edge?



**Surface orientation discontinuity**

# Image Features: Edges

- What causes an edge?



**Reflectance discontinuity  
(i.e. change of material)**

# Image Features: Edges

- What causes an edge?



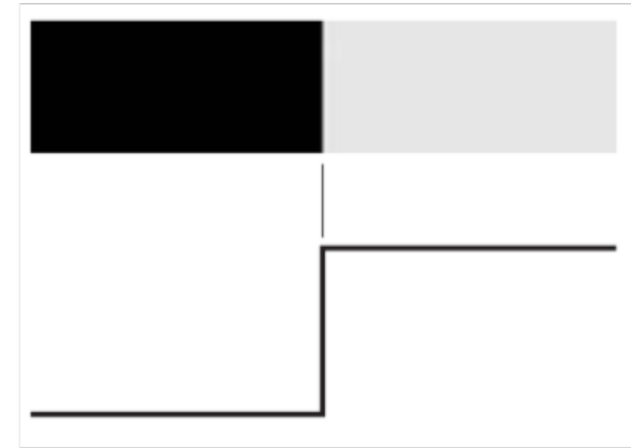
**Surface color discontinuity**

# Edge detection: Introduction

- Find image pixels that present abrupt (local) changes in intensity.

## Different edge models: Step Edges

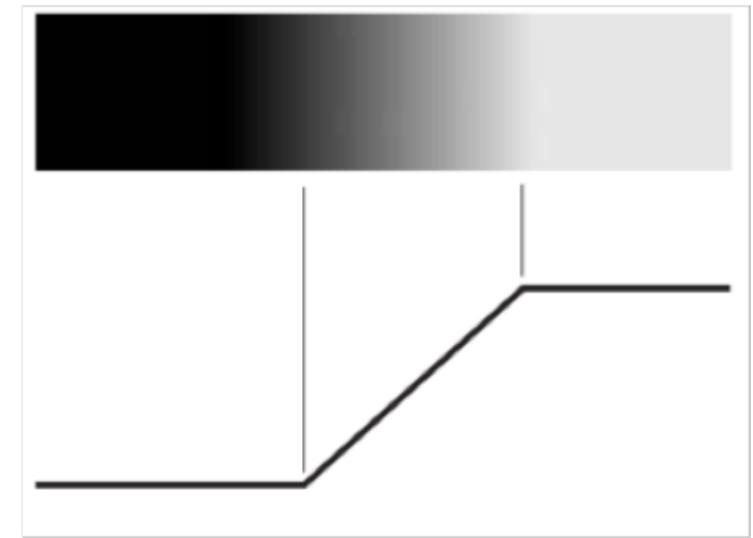
- **Description:** The intensity abruptly changes from one level to another across a boundary.
- **Model:** Ideal for modeling sharp, well-defined edges.
- **Mathematical shape:** A step function (Heaviside-like).
- **Examples:** Boundary between a black object and a white background.



# Edge detection: Introduction

## Different edge models: Ramp edges

- **Description:** The intensity change is gradual over a few pixels.
- **Model:** Represents real-world edges affected by blurring or gradual transitions.
- **Mathematical shape:** A smooth transition; often modeled as a linear or sigmoid ramp.
- **Examples:** Shadow boundaries or edges in out-of-focus regions.

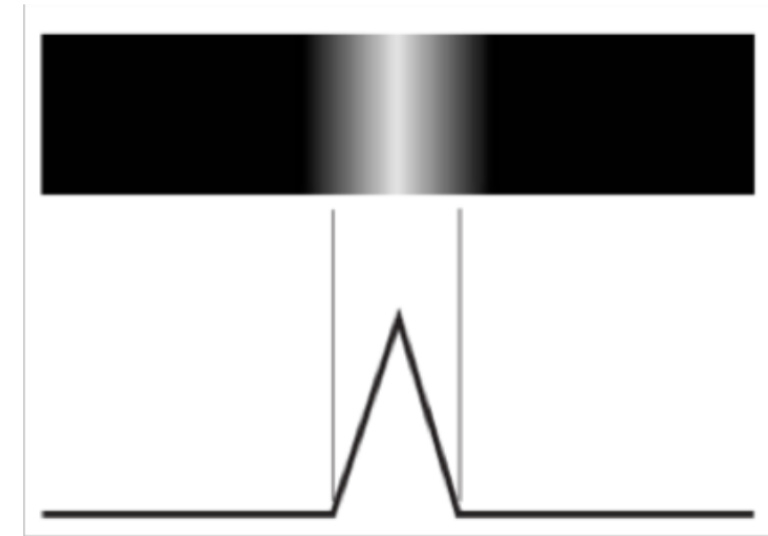


The slope of the ramp is inversely proportional to the degree of blurring in the edge.

# Edge detection: Introduction

## Different edge models: Roof edges

- **Description:** Intensity increases to a peak and then decreases symmetrically.
- **Model:** Represents edges formed by narrow ridges or lines.
- **Mathematical shape:** A peaked or triangular profile.
- **Examples:** Thin structures like wires, roads in aerial images, or fine hair.





# Edge detection: Introduction

**Edge models:** Noise result in deviations from the ideal shapes.



# Edge detection: Introduction

**An edge detector typically produces:**

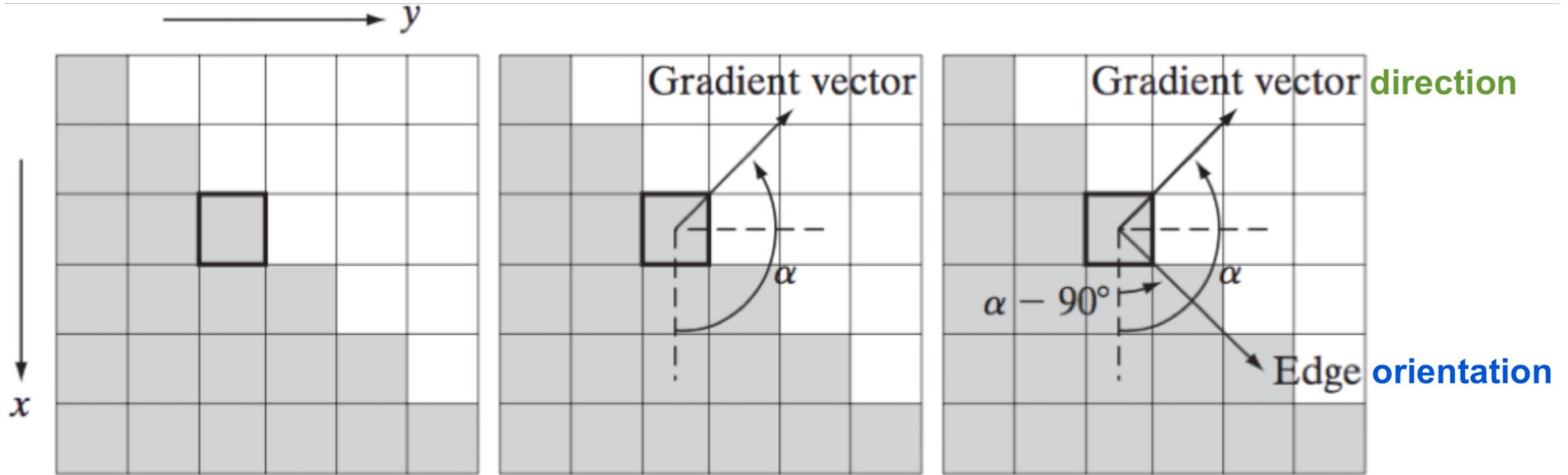
1. Edge **position**.
2. Edge **magnitude** (strength).
3. Edge **orientation** (orthogonal to the gradient **direction**).

**Performance requirements:**

1. High detection rate.
2. Good localization
3. Low sensitivity to noise.



# Edge detection: Introduction



- **Direction** of an edge refers to the way the intensity changes from one side of the edge to the other.
- **Orientation** describes the angle in which the edge is aligned.

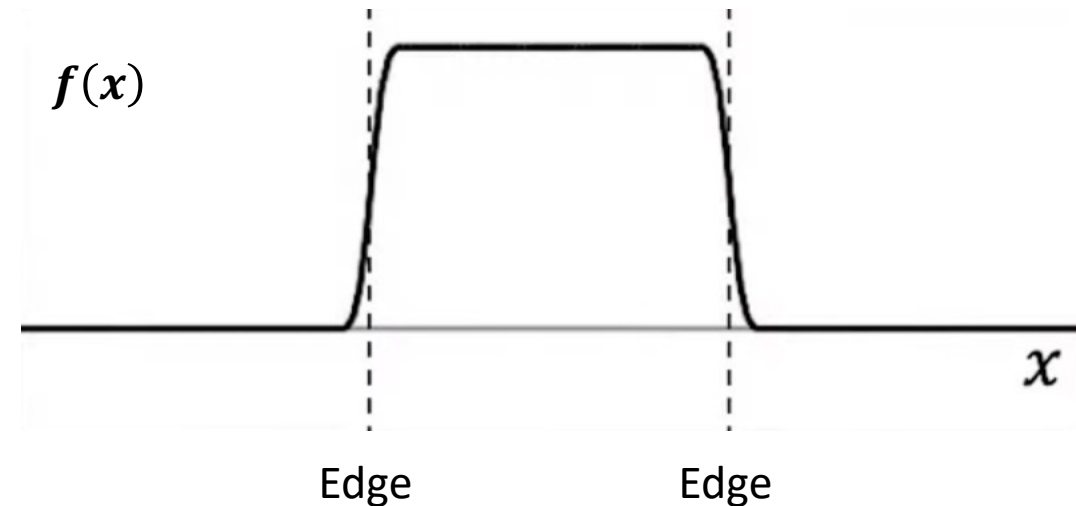
⇒ Direction is about the change in intensity while orientation is about the alignment of that change.

# I. Edge Detection using Gradients

- 1-D edge detection:

**Edge:** Rapid change in image intensity in a small region.

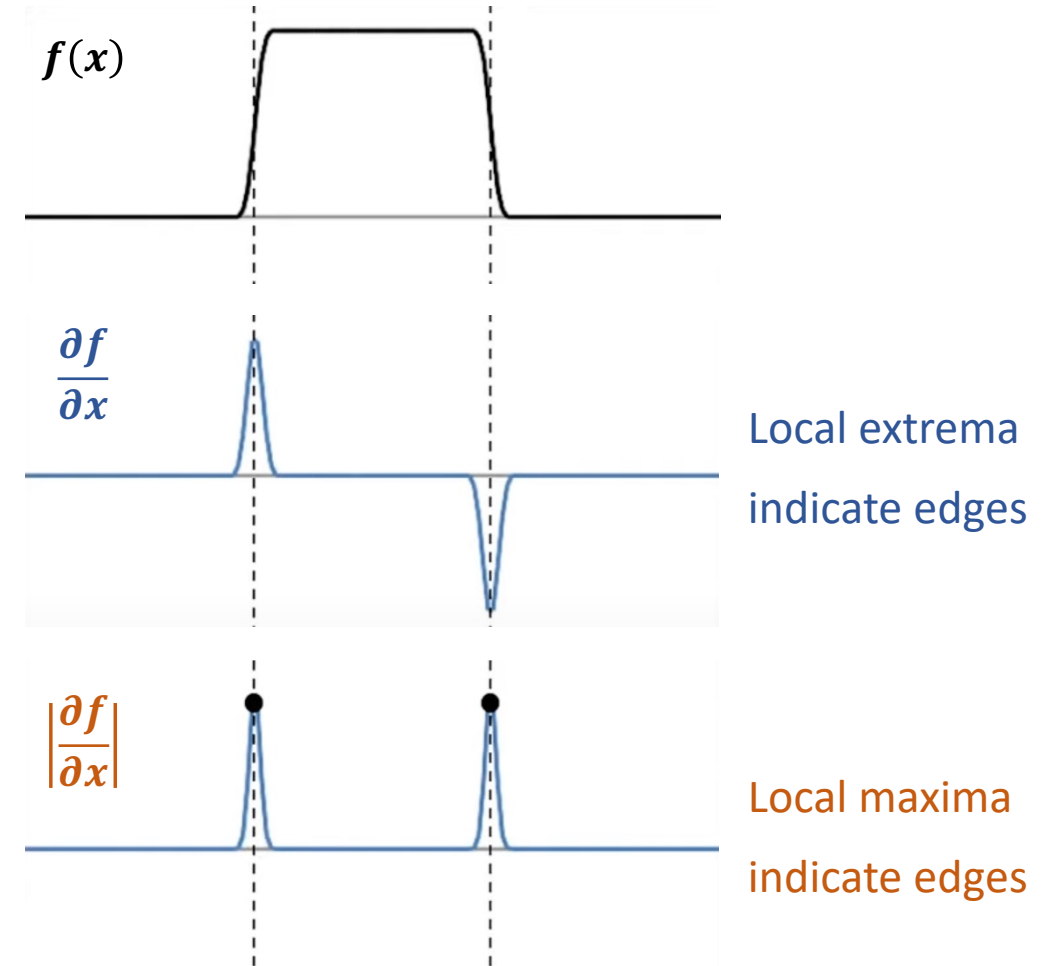
We know that the derivative of a continuous function represents the amount of change in the function.



- The **magnitude of the first derivative** can be used to detect the presence of an edge at a point in an image.

# I. Edge Detection using Gradients

- 1-D edge detection using derivatives

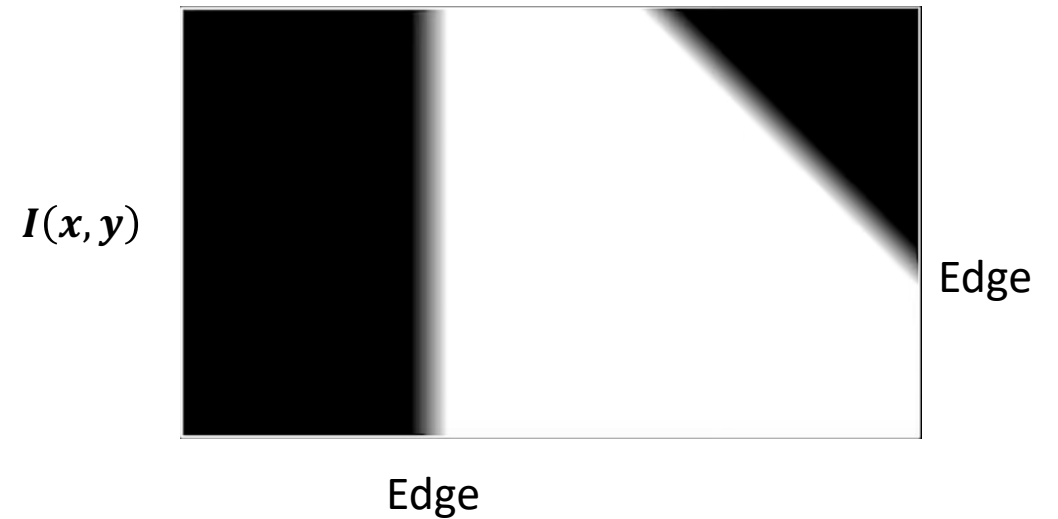


⇒ We get both location and strength of edges.

# I. Edge Detection using Gradients

- 2D edge detection using derivatives:

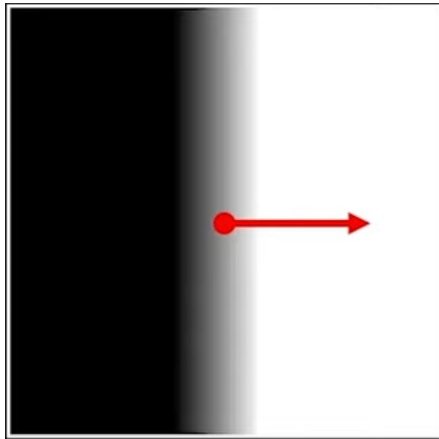
**Basic calculus:** Partial derivatives of a continuous 2D function represents the amount of change along each dimension.



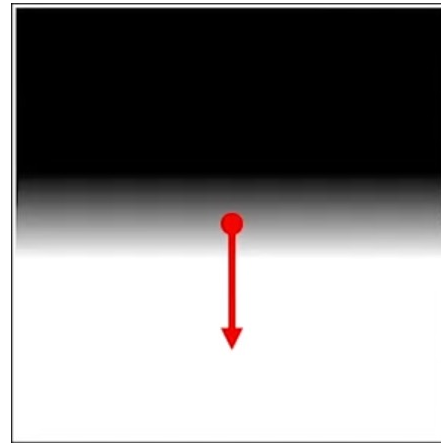
# I. Edge Detection using Gradients

- 2D edge detection using derivatives: The gradient operator ( $\nabla$ )

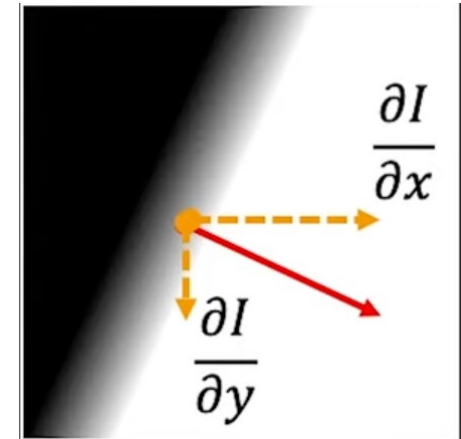
$$\nabla I = \left[ \frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right]$$



$$\nabla I = \left[ \frac{\partial I}{\partial x}, 0 \right]$$



$$\nabla I = \left[ 0, \frac{\partial I}{\partial y} \right]$$



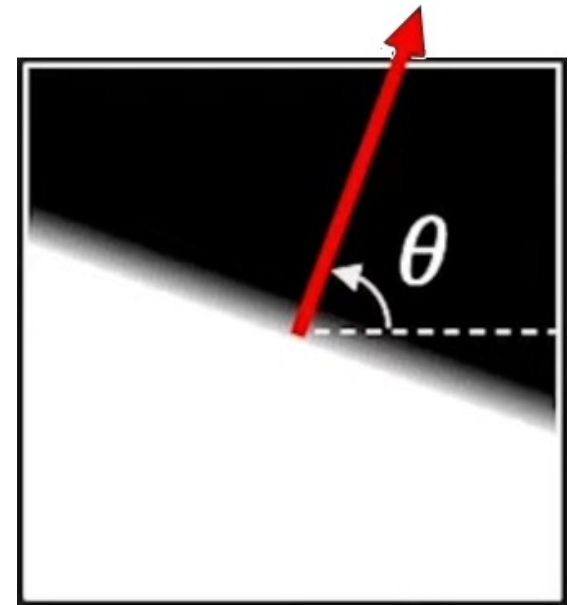
$$\nabla I = \left[ \frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right]$$

# I. Edge Detection using Gradients

- 2D edge detection using derivatives: The gradient operator ( $\nabla$ )

**Magnitude:**  $S = \|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2}$

**Direction:**  $\theta = \tan^{-1} \left( \frac{\frac{\partial I}{\partial y}}{\frac{\partial I}{\partial x}} \right)$

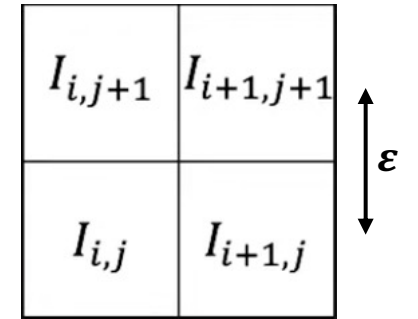


# I. Edge Detection using Gradients

- Application to discrete images:

$$\frac{\partial I}{\partial x} \approx \frac{1}{2\varepsilon} \left( (I_{i+1,j+1} - I_{i,j+1}) + (I_{i+1,j} - I_{i,j}) \right)$$

$$\frac{\partial I}{\partial y} \approx \frac{1}{2\varepsilon} \left( (I_{i+1,j+1} - I_{i+1,j}) + (I_{i,j+1} - I_{i,j}) \right)$$



- It can be implemented as convolution:

$$\frac{\partial I}{\partial x} \approx \frac{1}{2\varepsilon} \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\frac{\partial I}{\partial y} \approx \frac{1}{2\varepsilon} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

# I. Edge Detection using Gradients

- A variety of gradient operator has been proposed:

| Gradient                        | Roberts                                                                    | Prewitt | Sobel (3x3) | Sobel (5x5) |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
|---------------------------------|----------------------------------------------------------------------------|---------|-------------|-------------|----|-----------------------------------------------------------------------------------------------------------------------------------------|----|---|---|----|---|---|----|----|----|-----------------------------------------------------------------------------------------------------------------------------------------|----|---|---|----|---|---|----|----|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|---|----|----|---|---|---|----|----|---|---|---|----|----|----|----|----|----|----|----|----|----|
| $\frac{\partial I}{\partial x}$ | <table><tr><td>0</td><td>1</td></tr><tr><td>-1</td><td>0</td></tr></table> | 0       | 1           | -1          | 0  | <table><tr><td>-1</td><td>0</td><td>1</td></tr><tr><td>-1</td><td>0</td><td>1</td></tr><tr><td>-1</td><td>0</td><td>1</td></tr></table> | -1 | 0 | 1 | -1 | 0 | 1 | -1 | 0  | 1  | <table><tr><td>-1</td><td>0</td><td>1</td></tr><tr><td>-2</td><td>0</td><td>2</td></tr><tr><td>-1</td><td>0</td><td>1</td></tr></table> | -1 | 0 | 1 | -2 | 0 | 2 | -1 | 0  | 1  | <table><tr><td>-1</td><td>-2</td><td>0</td><td>2</td><td>1</td></tr><tr><td>-2</td><td>-3</td><td>0</td><td>3</td><td>2</td></tr><tr><td>-3</td><td>-5</td><td>0</td><td>5</td><td>3</td></tr><tr><td>-2</td><td>-3</td><td>0</td><td>3</td><td>2</td></tr><tr><td>-1</td><td>-2</td><td>0</td><td>2</td><td>1</td></tr></table> | -1 | -2 | 0 | 2 | 1 | -2 | -3 | 0 | 3 | 2 | -3 | -5 | 0 | 5 | 3 | -2 | -3 | 0  | 3  | 2  | -1 | -2 | 0  | 2  | 1  |
| 0                               | 1                                                                          |         |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | 0                                                                          |         |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | 0                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | 0                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | 0                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | 0                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -2                              | 0                                                                          | 2       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | 0                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | -2                                                                         | 0       | 2           | 1           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -2                              | -3                                                                         | 0       | 3           | 2           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -3                              | -5                                                                         | 0       | 5           | 3           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -2                              | -3                                                                         | 0       | 3           | 2           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | -2                                                                         | 0       | 2           | 1           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| $\frac{\partial I}{\partial y}$ | <table><tr><td>1</td><td>0</td></tr><tr><td>0</td><td>-1</td></tr></table> | 1       | 0           | 0           | -1 | <table><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>-1</td><td>-1</td><td>-1</td></tr></table> | 1  | 1 | 1 | 0  | 0 | 0 | -1 | -1 | -1 | <table><tr><td>1</td><td>2</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>-1</td><td>-2</td><td>-1</td></tr></table> | 1  | 2 | 1 | 0  | 0 | 0 | -1 | -2 | -1 | <table><tr><td>1</td><td>2</td><td>3</td><td>2</td><td>1</td></tr><tr><td>2</td><td>3</td><td>5</td><td>3</td><td>2</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>-2</td><td>-3</td><td>-5</td><td>-3</td><td>-2</td></tr><tr><td>-1</td><td>-2</td><td>-3</td><td>-2</td><td>-1</td></tr></table> | 1  | 2  | 3 | 2 | 1 | 2  | 3  | 5 | 3 | 2 | 0  | 0  | 0 | 0 | 0 | -2 | -3 | -5 | -3 | -2 | -1 | -2 | -3 | -2 | -1 |
| 1                               | 0                                                                          |         |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 0                               | -1                                                                         |         |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 1                               | 1                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 0                               | 0                                                                          | 0       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | -1                                                                         | -1      |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 1                               | 2                                                                          | 1       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 0                               | 0                                                                          | 0       |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | -2                                                                         | -1      |             |             |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 1                               | 2                                                                          | 3       | 2           | 1           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 2                               | 3                                                                          | 5       | 3           | 2           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| 0                               | 0                                                                          | 0       | 0           | 0           |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -2                              | -3                                                                         | -5      | -3          | -2          |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |
| -1                              | -2                                                                         | -3      | -2          | -1          |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                         |    |   |   |    |   |   |    |    |    |                                                                                                                                                                                                                                                                                                                                  |    |    |   |   |   |    |    |   |   |   |    |    |   |   |   |    |    |    |    |    |    |    |    |    |    |

Good localization



Poor localization

Noise sensitive



Less noise sensitive

Poor detection



Good detection



# I. Edge Detection using Gradients



Image  $I$



$$\frac{\partial I}{\partial x}$$



Gradient  
magnitude



$$\frac{\partial I}{\partial y}$$

# I. Edge Detection using Gradients

- Gradient using a 3x3 Sobel operator



Image  $I$



$$\frac{\partial I}{\partial y}$$



Gradient  
magnitude



$$\frac{\partial I}{\partial x}$$

# I. Edge Detection using Gradients

**Thresholding: Deciding which pixels definitely belong to an edge.**

- Standard: Using a single threshold ( $T$ )

$$\|\nabla I(x, y)\| < T$$

Definitely not an edge

$$\|\nabla I(x, y)\| \geq T$$

Definitely an edge

- Hysteresis-based: Using two thresholds ( $T_0 < T_1$ )

$$\|\nabla I(x, y)\| < T_0$$

Definitely not an edge

$$\|\nabla I(x, y)\| \geq T_1$$

Definitely an edge

$$T_0 \leq \|\nabla I(x, y)\| < T_1$$

Is an edge if a neighboring pixel is definitely an edge

# I. Edge Detection using Gradients

**Thresholding: Deciding which pixels definitely belong to an edge.**

- If the threshold is too high, important edges may be missed.
- If the threshold is too low, noise may be mistaken for a genuine edge.

# I. Edge Detection using Gradients

- Another example.



Image  $I$



$$\frac{\partial I}{\partial x}$$



Gradient  
magnitude



$$\frac{\partial I}{\partial y}$$

# I. Edge Detection using Gradients

- The Lena example.



Image  $I$



$$\frac{\partial I}{\partial x}$$



$$\frac{\partial I}{\partial y}$$



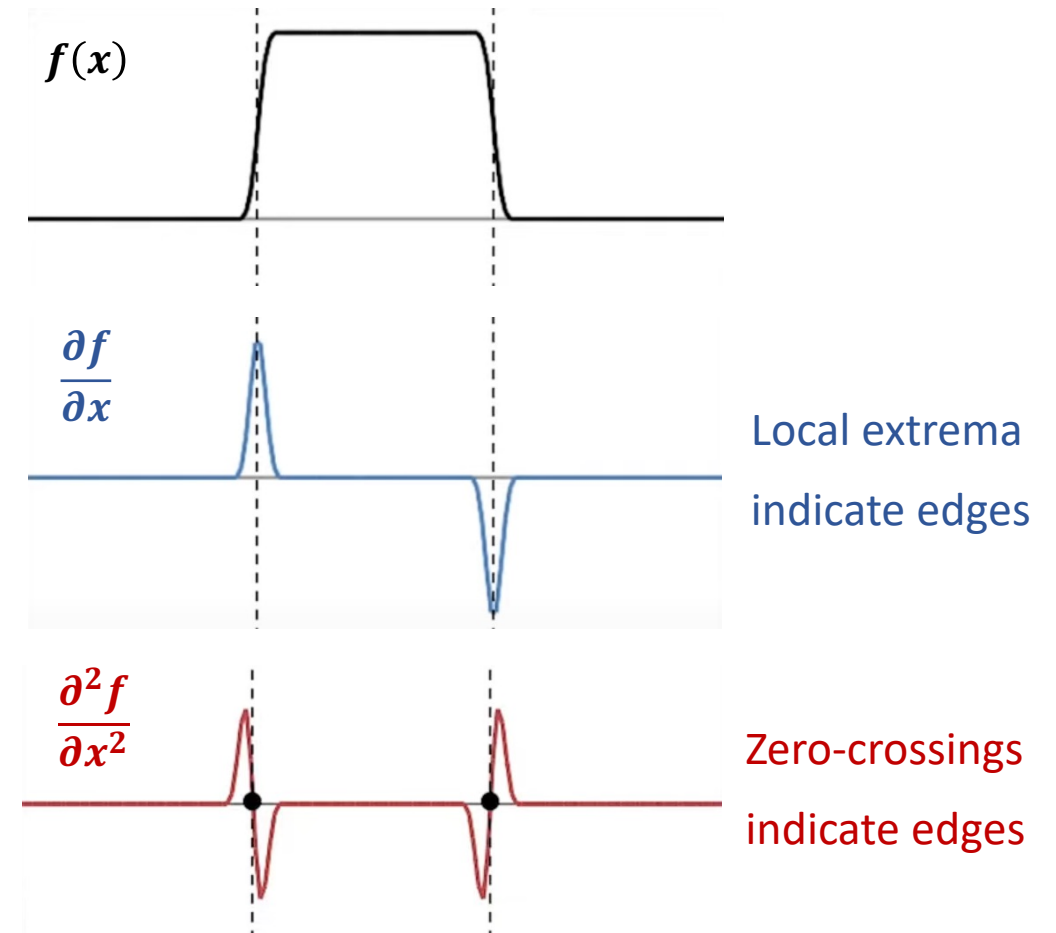
Gradient  
magnitude



Threshold  
edge

## II. Edge Detection using Laplacian

- Using the second derivative





## II. Edge Detection using Laplacian

- Laplacian ( $\nabla^2$ ) as edge detector

Laplacian: The sum of pure second derivatives.

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

This measures how much the intensity differs from its neighbors in **all directions**.

⇒ Edges are **zero-crossings** in the Laplacian image.



## II. Edge Detection using Laplacian

- Laplacian ( $\nabla^2$ ) as edge detector

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

- ⇒ A **zero-crossing** is a point where the **sign of the Laplacian response changes**: from positive to negative or vice versa.
- ⇒ These sign changes usually occur at **edges**, especially where there is a sharp change in intensity.
- ⇒ **Laplacian does not provide the direction of edges.**

## II. Edge Detection using Laplacian

- Discrete Laplacian operator ( $\nabla^2$ )

$$\frac{\partial^2 I}{\partial x^2} \approx \frac{1}{\varepsilon^2} (I_{i-1,j} - 2I_{i,j} + I_{i+1,j})$$

$$\frac{\partial^2 I}{\partial y^2} \approx \frac{1}{\varepsilon^2} (I_{i,j-1} - 2I_{i,j} + I_{i,j+1})$$

$$\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$$

|               |             |               |
|---------------|-------------|---------------|
| $I_{i-1,j+1}$ | $I_{i,j+1}$ | $I_{i+1,j+1}$ |
| $I_{i-1,j}$   | $I_{i,j}$   | $I_{i+1,j}$   |
| $I_{i-1,j-1}$ | $I_{i,j-1}$ | $I_{i+1,j-1}$ |

$\varepsilon$

- Using convolution:

$$\nabla^2 \approx \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\nabla^2 \approx \frac{1}{6} \begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix}$$

$$\nabla^2 \approx \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

## II. Edge Detection using Laplacian

### Step-by-step: How zero-crossings are detected

#### 1. Apply the Laplacian operator

- Convolve the image with a Laplacian kernel.
- This results in a **new 2D image** of Laplacian values: some positive, some negative.

#### 2. Check each pixel and its neighbors

To detect a zero-crossing at a pixel, we have to check **its 8-connected neighbors** (horizontal, vertical, and diagonal):

- If **the sign of the Laplacian value changes** between a pixel and one of its neighbors,
- AND the **absolute difference** is large enough (to avoid tiny sign flips due to noise),  
=> Then a **zero-crossing** (i.e., an edge) is detected.

## II. Edge Detection using Laplacian

- Application to the Lena image

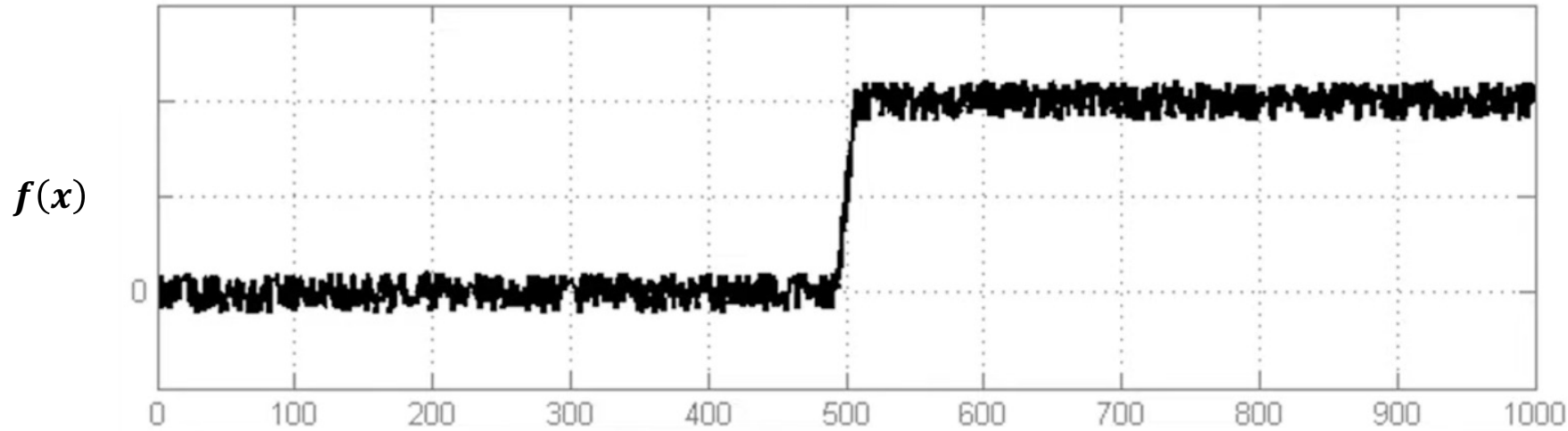


Laplacian  
(0 maps to 128)

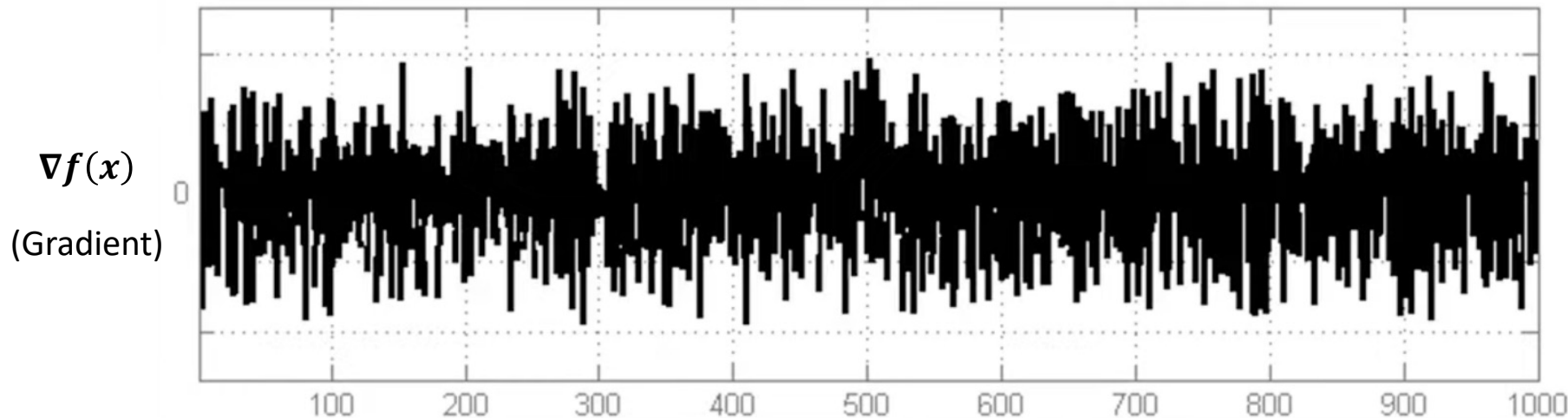


Laplacian  
Zero-crossings

# III. Edge Detection: Derivatives and Noise



*The noise introduces rapid changes in the signal – and the edge is also defined as a rapid change in the signal*



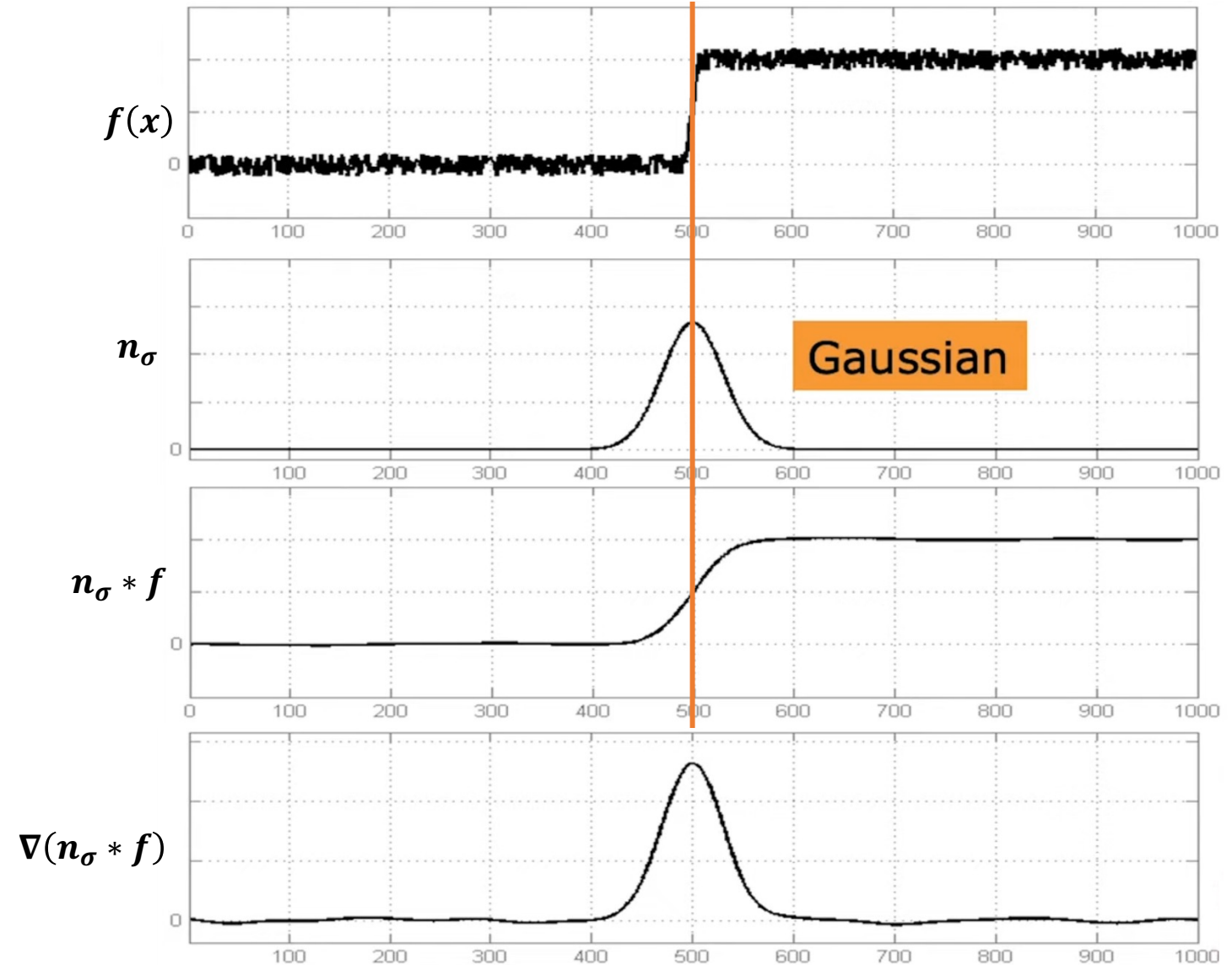
*Noise leads to a **false interpretation** of the edge.  
The edge is completely lost.*

### III. Edge Detection: Derivatives and Noise

- Noise is an important issue to keep in mind since it can have a significant impact on the two key derivatives used for detecting edges.
  - The noise has to be removed/mitigated before applying edge detection.
- ⇒ **The solution:** Image smoothing. It is usually a mandatory step prior to the use of derivatives.

### III. Edge Detection: Derivatives and Noise

- Smoothing the signal first using Gaussian filtering



# III. Edge Detection: Derivatives and Noise

## Using derivate of Gaussian

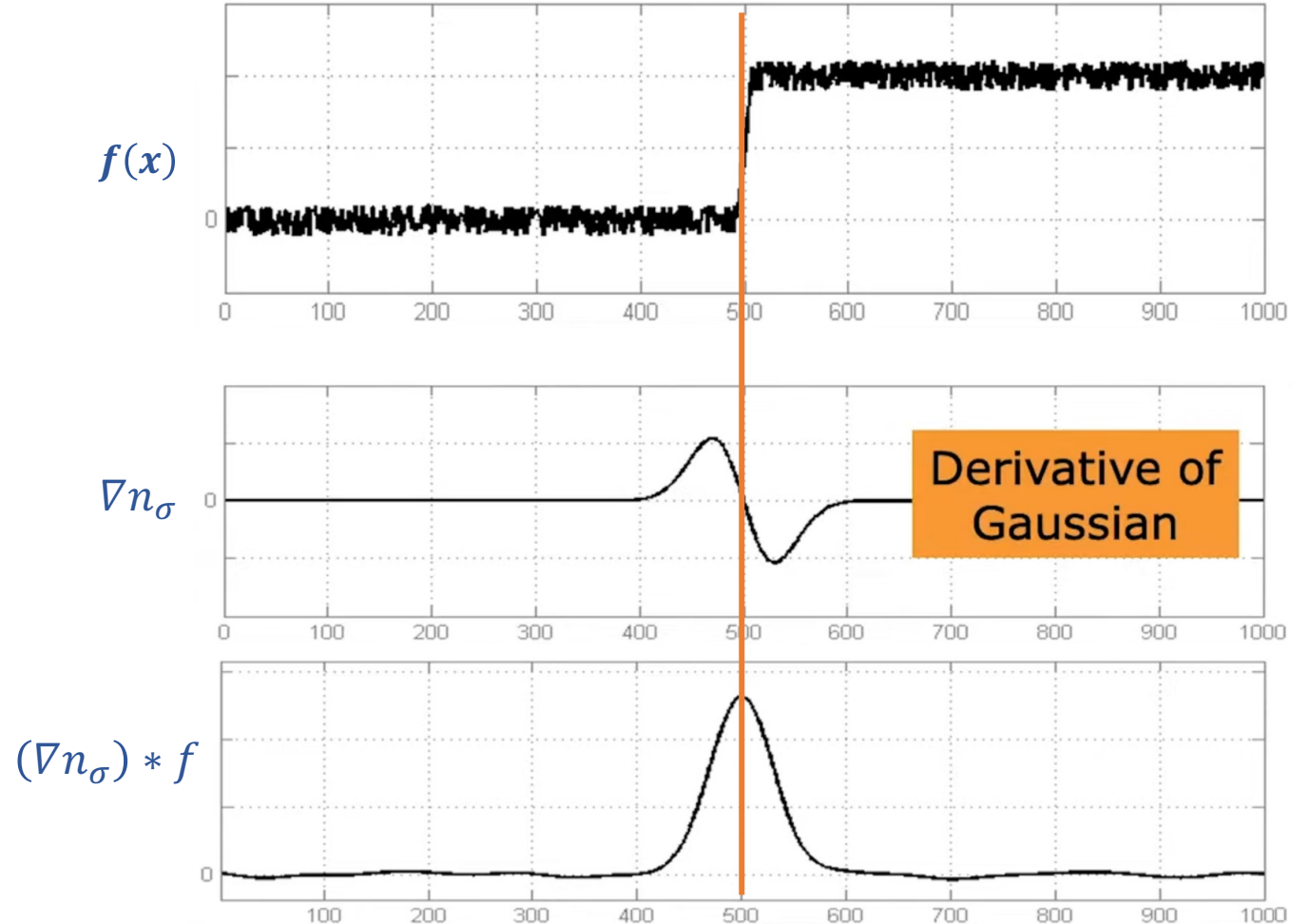
- We use  $\nabla(n_\sigma * f)$
  - We know that:
    - $\nabla$  is a linear operation
    - Gaussian smoothing is linear
- $\Rightarrow \nabla(n_\sigma * f) = (\nabla n_\sigma) * f$
- $\Rightarrow \nabla n_\sigma$  is a new operator that can be computed once and convolved with  $f$  to get the derivative on a smoothed version of  $f$



# III. Edge Detection: Derivatives and Noise

## Using derivate of Gaussian

$$\nabla(n_\sigma * f) = (\nabla n_\sigma) * f$$



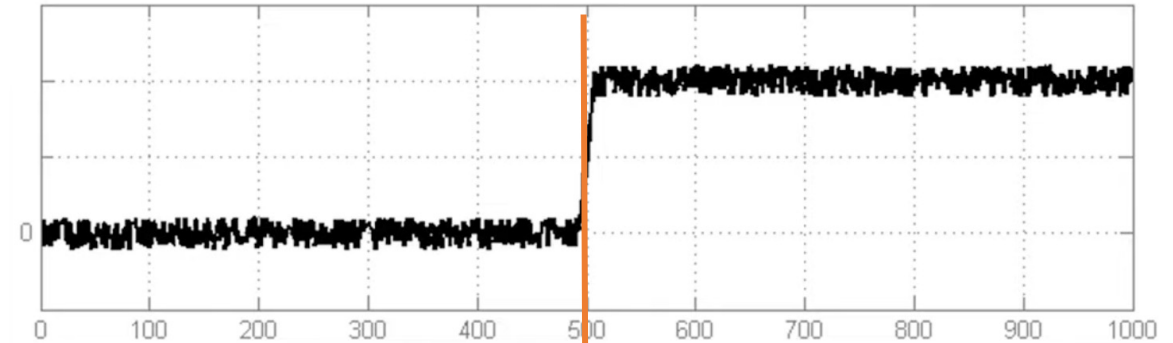
# III. Edge Detection: Derivatives and Noise

- The same can be applied to Laplacian:

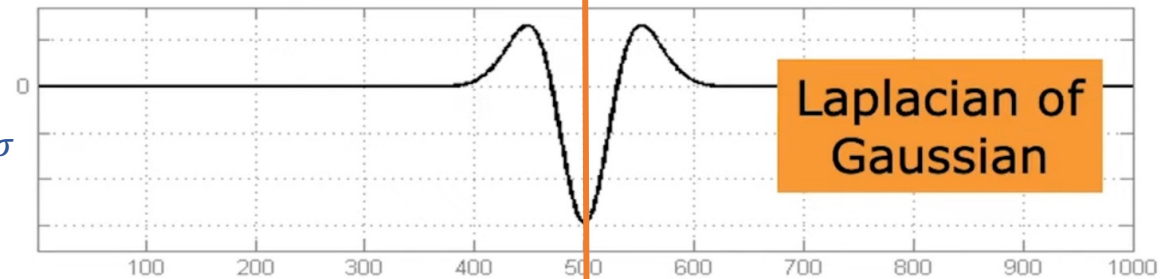
**Laplacian of Gaussian ( $\nabla^2 G$ )**

$$\nabla^2(n_\sigma * f) = (\nabla^2 n_\sigma) * f$$

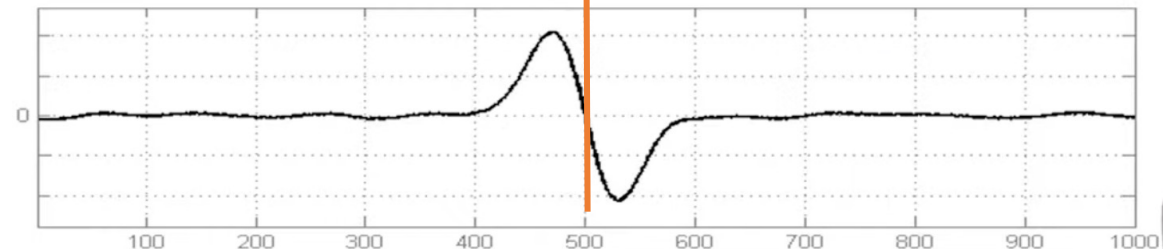
$f(x)$



$\nabla^2 n_\sigma$



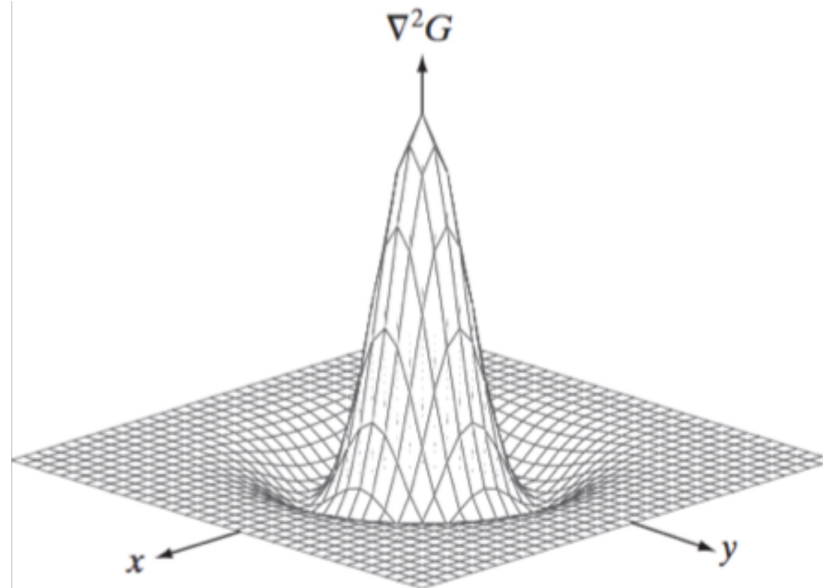
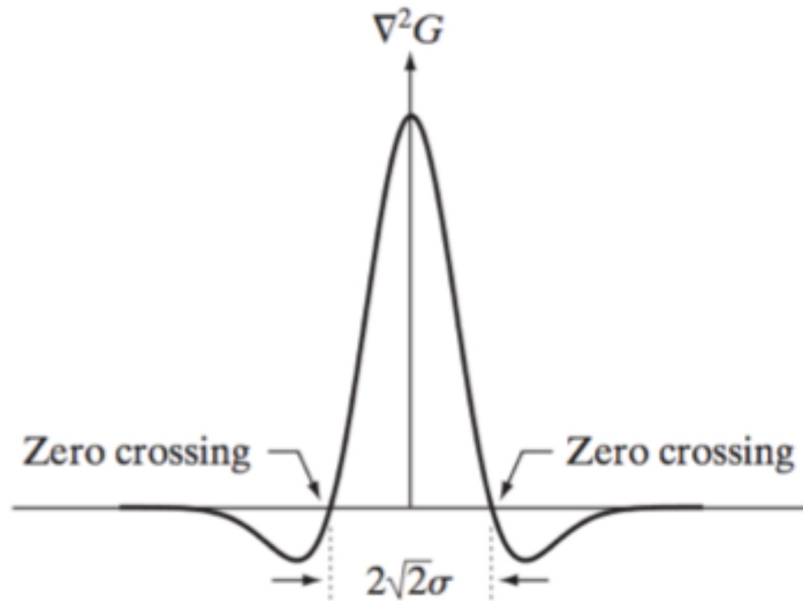
$(\nabla^2 n_\sigma) * f$



# III. Edge Detection: Derivatives and Noise

- Laplacian of Gaussian ( $\nabla^2 G$ )

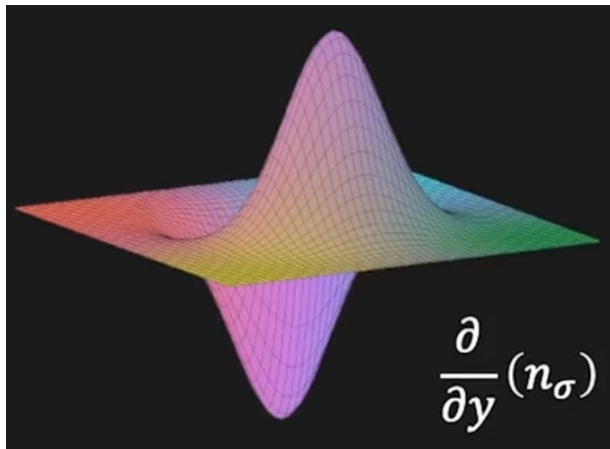
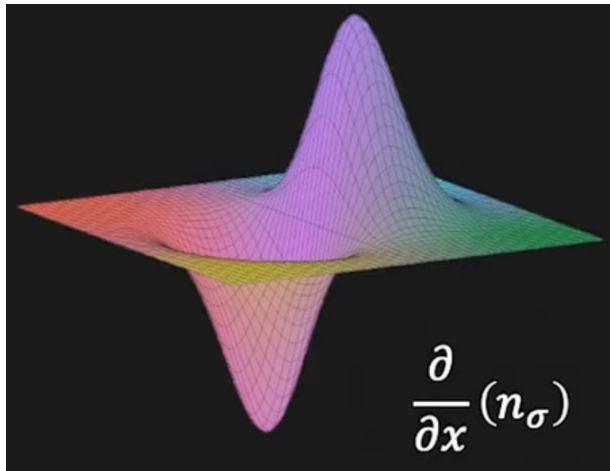
$$\begin{aligned}\nabla^2 G(x, y) &= \frac{\partial e^{-\frac{x^2+y^2}{2\sigma^2}}}{\partial x^2} + \frac{\partial e^{-\frac{x^2+y^2}{2\sigma^2}}}{\partial y^2} \\ &= \left[ \frac{x^2 + y^2 + 2\sigma^2}{\sigma^4} \right] e^{-\frac{x^2+y^2}{2\sigma^2}}\end{aligned}$$



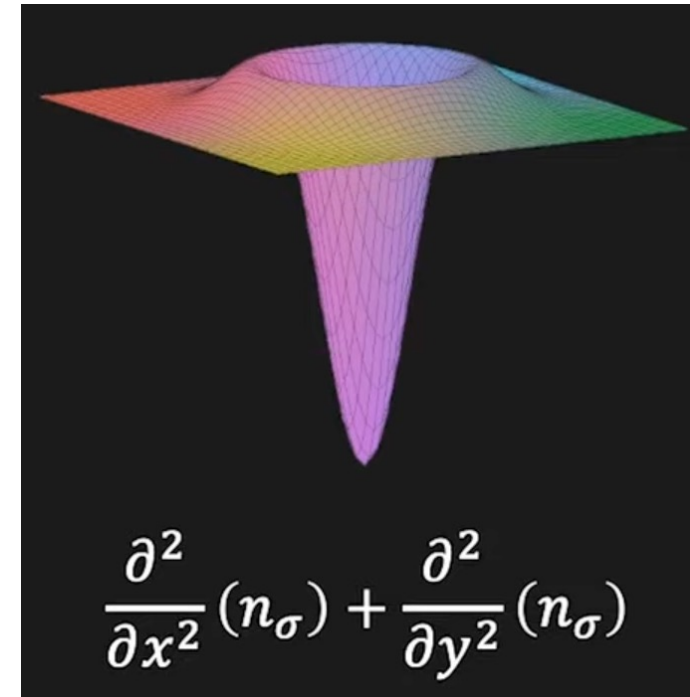
|    |    |    |    |    |
|----|----|----|----|----|
| 0  | 0  | -1 | 0  | 0  |
| 0  | -1 | -2 | -1 | 0  |
| -1 | -2 | 16 | -2 | -1 |
| 0  | -1 | -2 | -1 | 0  |
| 0  | 0  | -1 | 0  | 0  |

# III. Edge Detection: Gradient vs. Laplacian

**Gradient**



**Laplacian**



Inverted “sombrero”  
(Mexican hat)

# III. Edge Detection: Gradient vs. Laplacian

## Gradient

- Provide location, magnitude and orientation.
- Detection using maxima thresholding.
- Non-linear operation, requires two convolutions.

## Laplacian

- Provide only location of the edge.
- Detection based on zero-crossings.
- Linear operation, requires only one convolution.